INCH-POUND

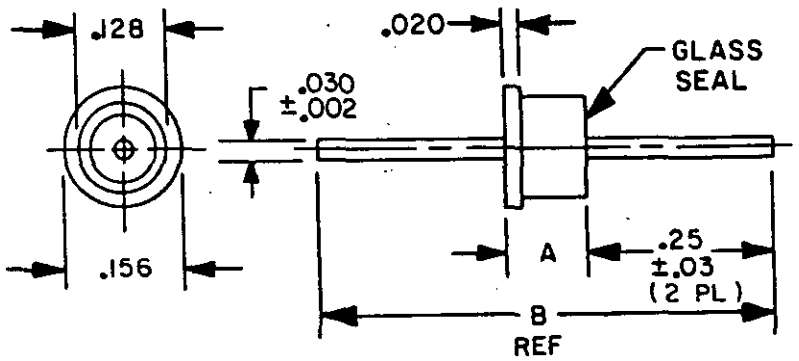
MIL-F-28861/12C
2 November 1988
SUPERSEDING
MIL-F-28861/12B
29 August 1986

MILITARY SPECIFICATION SHEET

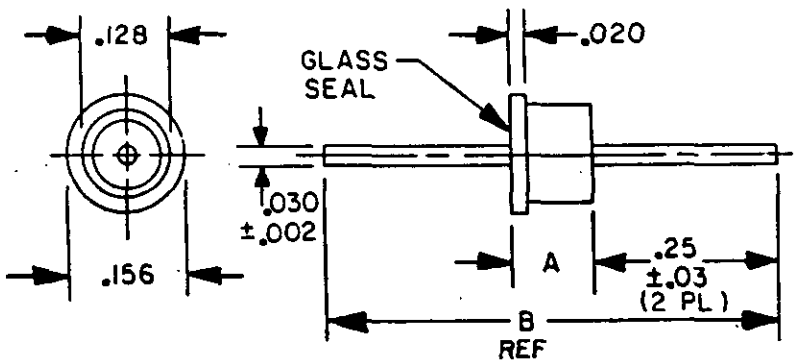
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED ON ONE END ONLY, STYLE FS70 AND FS71

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.



STYLE FS70



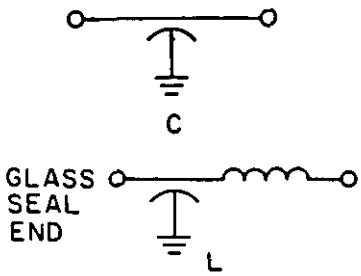
STYLE FS71

FIGURE 1. Case and hardware dimensions and circuit diagrams.

Ⓒ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CIRCUIT DIAGRAMS

Dash number	Style
001 thru 016, 033 and 034	FS70
017 thru 032, 035 and 036	FS71

Circuit diagram	Dimensions	
	A ±.005	B Ref
L	.200	.715
C	.110	.625

Inches	mm
.002	0.05
.005	0.13
.020	0.51
.030	0.76
.03	0.8
.110	2.79
.128	3.25
.156	3.96
.200	5.08
.25	6.4
.625	15.88
.715	18.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Circuit diagrams are for information only.
5. Filters shall be supplied with 60/40 solder preform.
6. Potting on nonhermetically sealed end shall not extend beyond .030 inch (0.76 mm) from the filter body.
7. Filters shall be installed using the recommended installation methods (solder-in style) of MIL-F-28861.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1. Filters and capacitors shall be hermetically sealed on one end as shown on figure 1 for the respective style.

Weight: .25 gram maximum for C circuit parts.
.75 gram maximum for L circuit parts.

Case and lead finish: G only (gold plated).

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 5 amperes maximum.

Capacitance: See table I.

Dissipation factor: .2 percent maximum for capacitance values 10 pF through 100 pF and 3 percent maximum for values greater than 100 pF.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm-microfarads or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarads or 10,000 megohms minimum, whichever is less.

Insertion loss:

At +25°C: In accordance with table I.

At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.05 volt maximum.

DC resistance: 0.01 ohm maximum.

Seal: Not applicable.

Temperature rise: +25°C maximum.

Solderability of terminals: In accordance with MIL-F-28861, except temperature of solder shall be 300°C ±5°C.

Resistance to soldering heat: In accordance with MIL-F-28861, except temperature of solder shall be 300°C ±5°C.

Solderability of mounting termination: In accordance with MIL-F-28861, except temperature of solder shall be 300°C ±5°C.

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© Quality assurance provisions:

Class B and class S filters: In accordance with MIL-F-28861.

Class A filters:

Qualification inspection: Initial qualification may be granted based on qualification to class B. Baseline documentation in accordance with MIL-F-28861 is required and shall be submitted to the qualifying activity prior to completion of quality conformance inspection on the initial lot. Retention of qualification shall be in accordance with class S requirements.

In-process inspection: In accordance with class S requirements, except that the requirements of MIL-C-123 shall not apply.

Quality conformance inspection: The inspection lot definition and inspection of product for delivery shall be in accordance with the class S requirements.

© TABLE I. Electrical characteristics.

Dash number	Circuit	Product assurance level			Rated voltage volts dc	Capacitance (pF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/				
		Class					1 MHz	10 MHz	100 MHz	1 GHz	10 GHz
		A	B	S							
001,017	L	---	X	---	50	15,000	7	25	45	60	60
002,018	C	---	X	---	50	15,000	7	25	40	60	60
003,019	L	X	X	X	100	2,700	---	10	25	50	60
004,020	C	X	X	X	100	2,700	---	10	25	50	60
005,021	L	X	X	X	100	5,000	---	15	30	50	60
006,022	C	X	X	X	100	5,000	---	15	30	50	60
007,023	L	X	X	X	200	10	---	---	---	15	20
008,024	C	X	X	X	200	10	---	---	---	5	20
009,025	L	X	X	X	200	25	---	---	---	20	25
010,026	C	X	X	X	200	25	---	---	---	10	25
011,027	L	X	X	X	200	100	---	---	3	20	40
012,028	C	X	X	X	200	100	---	---	3	20	40
013,029	L	X	X	X	200	500	---	---	15	34	55
014,030	C	X	X	X	200	500	---	---	15	34	50

See footnotes at end of table.

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© TABLE I. Electrical characteristics - Continued.

Dash number	Circuit	Product assurance level			Rated voltage volts dc	Capacitance (pF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/				
		Class					1 MHz	10 MHz	100 MHz	1 GHz	10 GHz
		A	B	S							
015,031	L	X	X	X	200	1,000	---	4	20	45	55
016,032	C	X	X	X	200	1,000	---	4	20	40	55
033,035	L	---	X	---	50	10,000	4	20	38	55	60
034,036	C	---	X	---	50	10,000	4	20	35	50	60

1/ For C circuits, insertion loss measurements shall be made under no load. For L circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz. Insertion loss measurements above this frequency range shall be made under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

© Product assurance level: In accordance with table I.

© Part or identifying number: The part or identifying number shall be as follows:

M28861/12 - XXX G X

- XXX: Dash no. per figure 1 and table I
- G: Case finish
- X: Product assurance level per MIL-F-28861 and table I

© Marking: Classes B and A capacitors and filters shall not be marked. Class S filters and capacitors shall be marked with a green dot in accordance with color number 14187 of FED-STD-595. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.

Cataloging information: Circuit 'C's shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Circuit 'L's shall be cataloged under FSC 5915 as radio frequency interference filters.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0074-01)

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